Adaptive Detector in Heterogeneous Clutter Based on Complex-Valued Convolutional Attention Network

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***Abstract*—To address the problem of target detection in heterogeneous clutter, a two-step adaptive detector is proposed in this paper. In the first step, a lightweight complex-valued convolutional attention network (CV-CoATNet) is designed to classify the clutter. In the second step, the data with the same distribution as the clutter in the cell under test are selected to design the generalized likelihood ratio detector. Simulation results show that the proposed adaptive detector based on CV-CoATNet achieves better detection performance than conventional adaptive detectors in heterogeneous clutter.**

***Index Terms*—Adaptive detection, Clutter classification, Complex-valued neural networks, Generalized likelihood ratio test**

I. Introduction

Adaptive signal detection is widely used in radar because it enables integrated clutter suppression and target detection through the design of a test statistic. Adaptive detection algorithms rely on statistical modeling of the received signals to estimate unknown parameters and dynamically adjust detection thresholds [1], [2]. Conventional detectors are derived under the assumption that clutter is homogeneous and follows the Gaussian probability distribution. However, in practical scenarios, clutter power exhibits significant variations across range cells due to geographical features such as mountains and land–sea boundaries. Under these heterogeneous clutter conditions, the detection performance of conventional adaptive detection methods derived in the homogeneous clutter degrades [3].

The partially homogeneous (PHM) and compound Gaussian (CGM) models are widely used to model heterogeneous clutter with varying power in radar signal processing. In the PHM model, the clutter power of the cell under test (CUT) and that of the secondary data differ only by an unknown constant scaling factor. A modified generalized likelihood ratio test (GLRT) detection method, exploiting multi-channel array data, is proposed in [4] based on a symmetrically spaced linear array. An alternating estimation GLRT, designed for adaptive multichannel processing under nonuniform clutter, is proposed in [5] to preserve constant false alarm rate (CFAR). In the CGM model, clutter is modeled as the product of a complex Gaussian speckle component x and a non-negative real-valued texture component $\sqrt{\tau}$. A heterogeneous detector dependent on the Clutter-to-Noise Ratio (CNR) is designed in [6], enabling adaptive detection in K-distributed backgrounds. A CFAR GLRT based on stochastic virtual signals to model mismatch is proposed in [7]. A Maximum likelihood estimation with projected gradient descent-based GLRT detectors for mismatched distributed targets is proposed in [8], achieving CFAR in both texture and covariance. However, in multi-clutter environments with different statistical distributions, these detectors suffer significant performance degradation. It is necessary to first classify the secondary data into homogeneous parts and then select secondary data that share the same statistical distribution as the CUT to design the adaptive detector.

Recently, deep learning has emerged as a powerful alternative by learning discriminative features directly from raw radar data. A metacognitive framework for classifying Gaussian, K-distributed, and Pareto-distributed clutter to enable adaptive CFAR operation is introduced in [9]. A complex-valued convolutional ResNet (CV-CRN) for classifying four types of clutter environments is proposed in [10], outperforming its real-valued counterpart with the same architecture. Nevertheless, radar signals are complex-valued, which real-valued networks cannot fully exploit due to phase loss; while preserving phase, complex-valued networks often entail higher computational complexity and parameter counts due to coupled operations between real and imaginary components.

In this paper, we propose a two-step adaptive detection framework. In the first step, a clutter classifier based on a lightweight Complex-Valued Convolutional Attention Network (CV-CoATNet), inspired by the real-valued CoAtNet (RV-CoATNet) architecture [11] is designed to partition the clutter into homogeneous parts. Unlike the CV-CRN in [10], which employs a standard residual architecture with high computational cost, our CV-CoATNet integrates depthwise separable convolutions and attention mechanisms to preserve the complex-valued signal structure while significantly reducing

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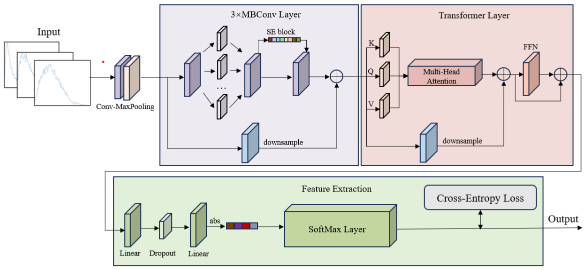


Fig. 1: Proposed CV-CoATNet Architecture

model complexity and inference latency. In the second step, the secondary data sharing the same statistical distribution as the CUT is selected to derive the GLRT. Simulation results validate the effectiveness of the proposed adaptive GLRT detector based on CV-CoATNet (CV-CoATNet-adapted).

The rest of this paper is organized as follows: Section II presents the proposed CV-CoATNet-adapted detector architecture. Simulated data was then used to test the proposed method in Section III. Finally, Section IV concludes the paper and outlines future research directions.

II. The Proposed Two-Step Detector

Radar clutter sequences are naturally represented as complex-valued signals. However, standard implementations of complex-valued neural networks double the parameter count compared to real-valued counterparts. To address these limitations, this paper proposes a lightweight CV-CoATNet to classify heterogeneous clutter. The network reduces complexity while preserving the geometric structure of complex radar signals.

A. Architecture of the CV-CoATNet

As illustrated in Fig. 1, the architecture of the CV-CoATNet consists of a complex convolution-pooling layer with a kernel size of 5, three Mobile Inverted Bottleneck Convolution (MBConv) blocks, a Transformer module to capture long-range dependencies, where rotary position embedding (RoPE) [12] is incorporated to encode sequential order into complex-valued features without altering their magnitude. CReLU is used as the activation function across all layers. Pre-activation residual connections [13] are used throughout to stabilize training:

$$\boldsymbol{x} \leftarrow \boldsymbol{x} + \text{Module}\big(\text{Norm}(\boldsymbol{x})\big), \tag{1}$$

where Norm denotes complex batch normalization (CBN) in MBConv blocks and complex layer normalization (CLN) in self-attention and FFN modules.

B. Complex-Valued Convolutional

The network first extracts local features and reduces the dimensionality of feature maps through convolution-pooling layers. For complex-valued input data $\boldsymbol{x} = \boldsymbol{a} + i\boldsymbol{b}$ and complex-valued weight vector $\boldsymbol{W} = \boldsymbol{A} + i\boldsymbol{B}$, the convolution operation is defined as [14]:

$$\boldsymbol{W} * \boldsymbol{x} = (\boldsymbol{A} * \boldsymbol{a} - \boldsymbol{B} * \boldsymbol{b}) + \mathrm{i}(\boldsymbol{B} * \boldsymbol{a} + \boldsymbol{A} * \boldsymbol{b}) \tag{2}$$

which can also be expressed in real-valued matrix form:

$$\begin{bmatrix} \Re(\boldsymbol{W} * \boldsymbol{x}) \\ \Im(\boldsymbol{W} * \boldsymbol{x}) \end{bmatrix} = \begin{bmatrix} \boldsymbol{A} & -\boldsymbol{B} \\ \boldsymbol{B} & \boldsymbol{A} \end{bmatrix} * \begin{bmatrix} \boldsymbol{a} \\ \boldsymbol{b} \end{bmatrix} \tag{3}$$

where $\boldsymbol{a}$, $\boldsymbol{b}$, $\boldsymbol{A}$ and $\boldsymbol{B}$ are all real-valued vectors, and $\Re(\bullet)$ and $\Im(\bullet)$ denote the real and imaginary parts.

C. Mobile Inverted Bottleneck Convolution

The "expand–depthwise–project" architecture of the MBConv block [15], [16] expands the channel dimension via pointwise convolution, extracts spatial features through depthwise convolution [17], and compresses the representation back to a lower-dimensional space via a projection layer. This design decouples spatial filtering from channel-wise transformations, significantly reducing computational cost and parameters while maintaining or even enhancing representational capacity.

An integrated Squeeze-and-Excitation (SE) channel attention module further adaptively recalibrates feature responses based on global context, selectively emphasizing more discriminative features—thereby improving the model's ability to distinguish complex clutter statistical patterns.

For downsampling within the MBConv module, a stride-2 convolution is applied to the normalized input on the residual branch:

$$\boldsymbol{x} \leftarrow \text{Proj}(\text{Pool}(\boldsymbol{x})) + \text{Conv}(\text{DepthConv}(\text{Conv}(\text{Norm}(\boldsymbol{x})))) \tag{4}$$

D. Rotary Position Embedding

To compensate for the insensitivity of the self-attention mechanism to input sequence order while preserving the geometric structure of complex-valued radar clutter, we employ RoPE. As illustrated in Fig. 2, the key idea is to design position-aware mappings f_q and f_k such that the inner product between query and key depends only on their relative offset [18]:

$$\langle f_q(\boldsymbol{x}_m, m), f_k(\boldsymbol{x}_n, n)\rangle = g(\boldsymbol{x}_m, \boldsymbol{x}_n, m-n), \tag{5}$$

where $\boldsymbol{x}_m, \boldsymbol{x}_n \in R_d$ are token embeddings, m, n are positions, and $g(\bullet)$ captures the dependence on relative position $m-n$.

For $d = 2$, RoPE leverages complex-plane rotation, wherein RoPE leverages the rotational geometry of 2D vectors in the complex plane to induce relative-position awareness:

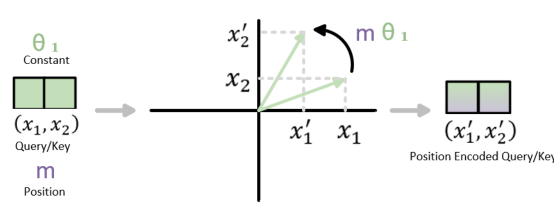


Fig. 2: Implementation of RoPE

$$
\begin{aligned}
f_q(\boldsymbol{x}_m, m) &= (\boldsymbol{W}_q \boldsymbol{x}_m) e^{im\theta} \\
f_k(\boldsymbol{x}_n, n) &= (\boldsymbol{W}_{\boldsymbol{k}} \boldsymbol{x}_n) e^{in\theta} \\
g(\boldsymbol{x}_m, \boldsymbol{x}_n, m-n) &= \Re\left[(\boldsymbol{W}_q \boldsymbol{x}_m)(\boldsymbol{W}_k \boldsymbol{x}_n)^* e^{i(m-n)\theta}\right]
\end{aligned} \tag{6}
$$

where $e^{i\theta}$ represents a rotation in the complex plane. This ensures the attention score depends only on the relative offset $m-n$, yielding translation-equivariant modeling. For $d \geq 2$, the embedding is split into consecutive 2D blocks, each rotated with frequency $\omega_k = 10000^{-2k/d}$.

E. Transformer

The Transformer employs a global attention mechanism to capture long-range dependencies across the sequence. This mechanism consists of three main components: CLN, multi-head attention (MHA), and a feed-forward network (FFN).

Given an input sequence $\boldsymbol{X} \in \mathbb{C}^{L \times M}$, we inject positional information via RoPE to obtain a position-aware representation $\tilde{X}$, based on which the multi-head attention with H heads is computed as:

$$
\boldsymbol{Q}_h = \tilde{\boldsymbol{X}} \boldsymbol{W}_Q^{(h)}, K_h = \tilde{\boldsymbol{X}} \boldsymbol{W}_K^{(h)}, V_h = \tilde{\boldsymbol{X}} \boldsymbol{W}_V^{(h)} \tag{7}
$$

$$
head_h = \text{softmax}\left(\frac{\boldsymbol{Q}_h \boldsymbol{K}_h^\top}{\sqrt{M/H}}\right) \boldsymbol{V}_h \tag{8}
$$

$$
\text{MHA}(\tilde{X}) = [head_1; \cdots; head_H] \boldsymbol{W}_O \tag{9}
$$

where $\boldsymbol{Q}_h$, $\boldsymbol{K}_h$, and $\boldsymbol{V}_h$ are query, key, and value projections, $d = M/H$ is the dimension of each head, and $\boldsymbol{W}_O$ is an output projection matrix.

For downsampling in the Transformer module, stride-2 max pooling is applied to the input of each branch within the self-attention module [19]:

$$
\boldsymbol{x} \leftarrow \text{Proj}(\text{Pool}(\boldsymbol{x})) + \text{Attention}(\text{Pool}(\text{Norm}(\boldsymbol{x}))) \tag{10}
$$

F. Output Layer

The output layer is implemented as a FFN comprising two fully connected layers with 64 and 4 neurons, respectively, interleaved with a dropout layer (rate = 0.5). To preserve the structural consistency of complex-valued features, the dropout mask is shared across the real and imaginary components at each position. The magnitude of the resulting 4-dimensional complex output vector is then computed and passed through a softmax layer to produce the clutter classification probabilities:

$$
FFN(\boldsymbol{x}) = \max(0, \boldsymbol{x}\boldsymbol{w}_1 + \boldsymbol{b}_1)\boldsymbol{w}_2 + \boldsymbol{b}_2 \tag{11}
$$

G. Adaptive Detector

After clutter classification, we select L homogeneous secondary data $\{\boldsymbol{z}_k\}_{k=1}^{L}$ classified by CV-CoATNet that share the same clutter statistical distribution as the CUT, formulate the target detection problem as a binary hypothesis test:

$$
\begin{cases}
\mathcal{H}_0 : \boldsymbol{z} = \boldsymbol{n}, \quad \boldsymbol{z}_k = \boldsymbol{n}_k, \quad k = 1, \ldots, L, \\
\mathcal{H}_1 : \boldsymbol{z} = \alpha\boldsymbol{p} + \boldsymbol{n}, \quad \boldsymbol{z}_k = \boldsymbol{n}_k, \quad k = 1, \ldots, L,
\end{cases} \tag{12}
$$

where $\boldsymbol{z}, \boldsymbol{z}_k \in \mathbb{C}^N$ denotes the data under test, $\boldsymbol{n}, \boldsymbol{n}_k \in \mathbb{C}^N$ denotes clutter, and $\alpha\boldsymbol{p}$ denotes the target with unknown complex amplitude α and steering vector $\boldsymbol{p}$, yielding the decision statistic:

$$
t_{\text{CV-CoATNet-adapted}} = \frac{\max\limits_{\boldsymbol{M},\alpha} \int f_{\mathcal{H}_1}(\boldsymbol{Z} \mid \boldsymbol{M}, \alpha, \tau) f(\tau) d\tau}{\max\limits_{\boldsymbol{M}} \int f_{\mathcal{H}_0}(\boldsymbol{Z} \mid \boldsymbol{M}, \tau) f(\tau) d\tau} \gtrless_{\mathcal{H}_0}^{\mathcal{H}_1} \eta \tag{13}
$$

where $f_{\mathcal{H}_1}(\boldsymbol{Z})$ and $f_{\mathcal{H}_0}(\boldsymbol{Z})$ denote the joint probability density functions under hypotheses $\mathcal{H}_1$ and $\mathcal{H}_0$, conditioned on the network output through the selected homogeneous set $\boldsymbol{Z} = \{\boldsymbol{z}, \boldsymbol{z}_1, \ldots, \boldsymbol{z}_L\}$, in which $\boldsymbol{M}$ denotes the clutter covariance matrix, α the unknown complex target amplitude, and $f(\tau)$ the PDF of the texture variable τ. η denotes the detection threshold.

III. Simulation Results and Analysis

A. Generation of Dataset

We employ the Zero-Memory Nonlinearity (ZMNL) method and the Spherically Invariant Random Process (SIRP) method [20] to simulate four types of clutter: Rayleigh, log-normal, Weibull, and K-distributed. The training set contains 10,000 samples per type, and the validation and test sets include 2,000 samples per type. Each range-bin sample consists of complex-valued data across 256 pulse dimensions, appended with a real-valued class label. To demonstrate robust generalization capability, a wide parameter range is adopted during simulation, enabling the model to learn more effective features over a broader operational region, as detailed in Table I. The validation set is strictly used for hyperparameter tuning to prevent data leakage during the final evaluation.

B. Network Training and Parameter Settings

Network hyperparameters are set as follows: a batch size of 64, a learning rate of 10^{-4}, and 50 training epochs. The Adam optimizer is employed with cross-entropy loss. Training and testing are conducted on a workstation equipped with an Intel Core i5-9300H processor@2.40 GHz and an NVIDIA GeForce GTX 1650 GPU.

TABLE I: Parameters for Clutter Distribution Simulation

Distribution Type	**Parameter**	**Range**	**Note**
Rayleigh	σ_v	0.1-5	σ_v: variance
Lognormal	μ	0.1-5	μ: mean
	σ	0.1-5	σ: standard deviation
Weibull	b	0.1-10	b: scale parameter
	a	0.1-10	a: shape parameter
K-distribution	v	0.1-10	v :shape parameter

TABLE II: Classification Accuracy under Different CNRs

Network	**Classification Accuracy (%)**			
	noise-free	**10 dB**	**5 dB**	**0 dB**
RV-CoATNet	96.75	93.13	88.75	83.54
CV-CRN	97.39	94.31	90.86	85.80
CV-Transformer	97.45	94.33	91.37	89.10
CV-CoATNet	97.52	94.50	93.16	90.08

TABLE III: Ablation Study of Network Components

Network	Params	Latency (ms)	Accuracy (%)
CV-CRN	597,912	0.87	97.39
CV-CRN+Transformer	1,389,336	0.93	97.45
CV-CRN+MBConv	251,816	0.56	97.30
CV-CoATNet	261,544	0.81	97.52

C. Clutter Classification Accuracy Under Different CNRs

To evaluate classification performance under different CNRs, the proposed CV-CoATNet is compared against RV-CoATNet, CV-CRN and complex-valued Transformer (CV-Transformer). Table II presents the classification results of the above four neural network architectures.

The comparison results reveal that complex-valued networks consistently outperform the RV-CoATNet in both accuracy and robustness, owing to their ability to exploit richer feature representations from the complex signal structure. While all methods degrade as noise intensity increases, CV-CoATNet achieves comparable accuracy to other complex models under noise-free conditions and demonstrates superior performance at low CNRs, indicating that its hybrid architecture enhanced noise resilience and overall robustness.

D. Ablation Study

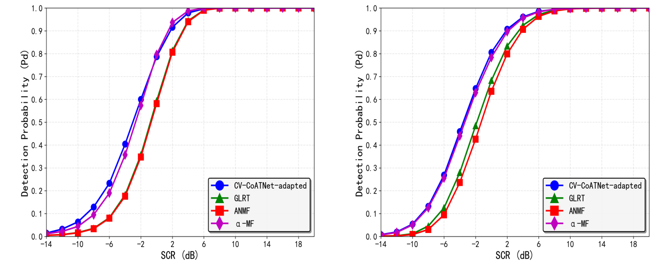
In this part, the effectiveness of the proposed hybrid architecture is validated through ablation studies (Table III). The integration of Transformer blocks improves accuracy, indicating that the global attention mechanism effectively enhances long-range dependency modeling. Conversely, incorporating lightweight MBConv modules substantially reduces the parameter count, demonstrating that its depthwise separable convolutions and SE channel attention optimize feature extraction while maintaining a compact footprint. In contrast, CV-CoATNet achieves the optimal balance by combining local convolutional feature extraction with the global receptive field of attention, thereby maximizing representational capacity.

E. Adaptive Radar Detection Based on Clutter Classification

In this part, the X-band sea clutter dataset ”20191012113102_01_staring” from the Journal of Radars [21] is used for validation. The dataset contains lognormal, K-distributed, and Rayleigh sea clutter, used to evaluate the detection performance of the proposed CV-CoATNet-adapted detector.

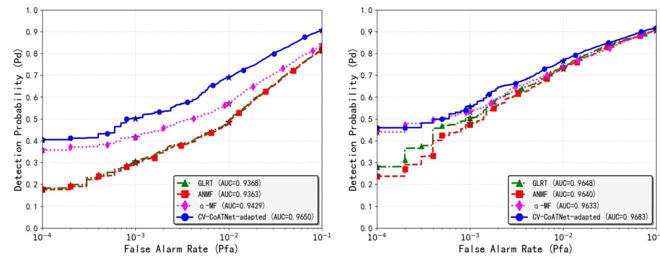
For comparison, the probabilities of detection of the GLRT [1], Adaptive Normalized Matched Filter (ANMF) [2] and α-MF [6], which do not perform clutter classification, are also given. As shown in Fig. 3b, the proposed CV-CoATNet-adapted detector achieves 1.57 dB and 1.65 dB detection performance gains compared with the GLRT and ANMF algorithms, while yielding performance comparable to the α-MF detector when N = 16, L = 64, Pfa = 10^{-4}.

Following the real-data analysis, Monte Carlo simulations are conducted under similar multi-clutter scenarios. As shown in Fig. 3a, the proposed CV-CoATNet-adapted detector



(a) Real-data results (b) Simulated results

Fig. 3: P_d versus SCR in heterogeneous clutter for N = 16, L = 64, $P_{\rm fa}$ = 10^{-4}



(a) Real-data results (b) Simulated results

Fig. 4: P_d versus $P_{\rm fa}$ of four detectors for N = 16, SCR = -4 dB

achieves 1.61 dB, 2.02 dB and 0.25 dB gains compared with the GLRT, ANMF and α-MF algorithms.

As shown in Fig. 4, the receiver operating characteristic (ROC) curves of the four detectors are presented for real and simulated scenarios at -4 dB signal-to-clutter ratio (SCR). The proposed CV-CoATNet-adapted detector achieves the best overall detection performance across all false alarm rates.

IV. Conclusion

In this paper, a two-step adaptive detector to detect targets in heterogeneous clutter is proposed. In the first step, a lightweight CV-CoATNet is designed for radar clutter classification. By combining depthwise separable convolutions with a streamlined self-attention mechanism, CV-CoATNet preserves the complex nature of radar echoes while achieving high efficiency. In the second step, the clutter selected by CV-CoATNet is used to design the GLRT detector. Experimental and Simulation results show that CV-CoATNet outperforms existing complex-valued models in accuracy and robustness under low CNR, with fewer parameters and lower latency. The proposed CV-CoATNet-adapted detector demonstrates superior performance compared to the GLRT, ANMF and α-MF algorithms in heterogeneous environments.

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